
EDUCATION
Doctor of Philosophy in Geography

University of California, Santa Barbara

- Spatial Conservation and Landscape Ecology (SCALE) Lab
- Graduate student affiliate, Center for Spatial Science

2024 - **Present****Master of Arts** in Geography

University of California, Santa Barbara

- Thesis: Spectrally-derived diversity estimates suggest coastal proximity as a driver of shrubland community composition in a threatened Mediterranean ecosystem

2024 - 2026

Bachelor of Science in Wildlife and Conservation Biology

University of New Hampshire

- Minor in Geospatial Analysis
- Dean's List Recipient: Fall 2021, Spring 2022, Fall 2022, Spring 2023

2019 - 2023

AWARDS AND FELLOWSHIPS

Pearl Chase Research Grant, UCSB \$7,200 2026

Geography Department Block Grant, UCSB \$4,000 2026

Graduate Student Scholarship, California Native Plant Society \$1,000 2026

Research Accelerator Award, Schmidt Family Foundation \$8,000 2025

Honorable Mention, NSF Graduate Research Fellowship Program 2025

Travel grant, IALE-NA BioLEAPS est. \$1,000 2025

Regents in Geography Fellow, UCSB est. \$68,000 2024 - 2028

Summer Undergraduate Research Fellow, UNH est. \$2,000 2022

PUBLICATIONS
Peer-reviewed:

Lopez, I. S., Fraser, B.T., & Congalton, R.G. (2023). Evaluating the Ability of Unpiloted Aerial Systems to Detect and Monitor Beech Bark Disease in New England Forests. *The Geographical Bulletin*, 64(2).
<https://digitalcommons.kennesaw.edu/thegeographicalbulletin/vol64/iss2/4>

Fraser, B.T., Bunyon, C.L., Reny, S., **Lopez, I. S.**, & Congalton, R.G. (2022). Analysis of Unmanned Aerial System (UAS) Sensor Data for Natural Resource Applications: A Review. *Geographies*, 2(2), Article 2. <https://doi.org/10.3390/geographies2020021>

Technical Reports:

X. Yan, Y. Zhan, W. Yang, O. Ross, Y. Liu, Z. Li, **I. Lopez**, I. Perez, E. Tao, J. Vogel, D. Cui, L. Song, A. Frazier (2025) Current and Future Habitat Suitability for Special Status Species at the Jack and Laura Dangermond Preserve. UC Santa Barbara: Center for Spatial Science. *EcoEvoRxiv* [Preprint]. <https://doi.org/10.32942/X2BH2N>

CONFERENCE PRESENTATIONS

Lopez, I., X. Yan, A.E. Frazier. (2026, April). Spectrally-derived diversity estimates indicate topography as a driver of shrubland diversity in a threatened Mediterranean ecosystem [Oral presentation]. *International Association of Landscape Ecologists (IALE) North America*, Athens, GA.

Lopez, I., X. Yan, A.E. Frazier. (2025, April). Relationships between community assemblage and coastal influence, elevation, and aspect: Spatial and spectral pattern analysis in a threatened Mediterranean ecosystem [Oral presentation]. *International Association of Landscape Ecologists (IALE) North America*, Raleigh, NC.

Frazier, A.E., D. Cui, L. Song, W. Yang, X. Yan, **I. Lopez**. (2025, April). Species on the Move: planning for future habitat shifts at the Dangermond Preserve under climate change [Oral presentation]. *Point Conception Institute Symposium*, Santa Barbara, CA.

Yan, X, A.E. Frazier, **I. Lopez**. (2025, March). Fire Effects on Plant Diversity: A Hyperspectral Analysis in California's Mediterranean Ecoregion [Oral presentation]. *Association of American Geographers (AAG) Annual Meeting*, Detroit, MI.

Lopez, I, B.T. Fraser, R.G. Congalton. (2023, April). The ability of Unmanned Aerial Systems to detect and monitor Beech Bark Disease [Oral presentation]. *Undergraduate Research Conference, UNH*, Durham, NH. **Awarded most outstanding oral presentation.**

Lopez, I, B.T. Fraser, R.G. Congalton. (2023, April). The ability of Unmanned Aerial Systems to detect and monitor Beech Bark Disease [Poster presentation]. *Undergraduate Research Conference, UNH*, Durham, NH.

TEACHING EXPERIENCE

Teaching Assistant— University of California, Santa Barbara

GEOG 115A Remote Sensing of the Environment I (Instructor: Joe McFadden)	Fall 2026
GEOG 115C Remote Sensing of the Environment III (Instructor: Zoe Pierrat)	Spring 2026
GEOG 115B Remote Sensing of the Environment II (Instructor: Amy Frazier)	Winter 2026
EARTH 7 Dinosaurs (Instructor: Andy Wyss)	Fall 2025

Guest Lecturer— University of California, Santa Barbara

GEOG 189/289 Close-Range Remote Sensing	Fall 2025
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SERVICE AND COMMUNITY ENGAGEMENT

Vice Chair, Graduate Application Materials Assistance Program, UCSB Geography	2026 - Present
Roles Lead, Committee on Committees, UCSB Geography	2026 - Present
Member, California Native Plant Society Channel Islands Chapter Conservation Committee	2026 - Present
Chair , Graduate Application Materials Assistance Program, UCSB Geography	2025 - 2026
Reviewer, Graduate Application Materials Assistance Program, UCSB Geography	2024 - Present
Graduate Student Association Representative, UCSB Geography	2024 - 2025
Undergraduate Representative, Diversity, Equity, and Inclusion Committee, UNH	2021 - 2023

Independent study mentor:

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| • Adele Wrench, California Polytechnic University | 2026 |
| • Schuyler Capita, UC Santa Barbara | 2024 - 2026 |

Geoscience Enrichment and Mentorship for Students by Students (GEMSS) mentor:

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| • Lily Strange, UC Santa Barbara | 2024 - 2026 |
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Professional memberships:

- California Native Plant Society
- Southern California Botanists
- International Association of Landscape Ecologists (IALE) North America

RELEVANT RESEARCH & PROFESSIONAL EXPERIENCE

Bren School of Environmental Science & Management — UC Santa Barbara, CA	September - November 2025
<i>Bren Environmental Leadership Fellow</i>	

- Served as a graduate student leader in The Nature Conservancy's Environmental Education Program
- Led groups of high school students on guided educational hikes at the Jack and Laura Dangermond Preserve

NH Fish & Game: Nongame & Endangered Wildlife Division — Concord, NH

April - August 2024

Karner Blue Butterfly Biological Technician

- Worked with a team to captive rear over 150 federally endangered Karner blue butterflies at all life stages in a lab
- Participated in restoration and habitat management of pine barrens habitat through intensive field work
- Monitored populations of sundial lupine and Karner blue butterflies through count transects and census surveys
- Served as point-person in USFWS-funded Monarch butterfly DNA collection project, coordinated and oversaw humane collection of over 100 Monarch caterpillar individuals across 15 sites throughout New Hampshire
- Used ArcGIS Online and iNaturalist data to identify public lands with Monarchs and milkweed populations throughout New Hampshire

The Nature Conservancy — Newmarket, NH

May - December 2023

Seasonal Land Steward & Volunteer Coordinator

- Served as a field crew leader to two Americorps members, oversaw stewardship work as well as planned and coordinated weekly stewardship tasks and assessed performance through performance reviews
- Conducted property inspections for 30 TNC preserves across New Hampshire to identify ecological issues, management problems, and evaluate conservation values
- Worked with volunteers on a weekly basis as well as organized volunteering events, including invasive plant removals and trail bridge implementations
- Earned wildland firefighter certification via 35 hours of trainings and passed moderate Pack Test in order to participate in fire line preparation, prescribed burns, and mop-up to improve pine barren ecology

EcoQuest Center for Indigeneity, Ecology and Creativity — Aotearoa, New Zealand

September - December 2022

Student Researcher

- Selected for competitive semester abroad for a hands-on applied field research experience
- Researched wētā (Orthoptera: Anostomatidae and Rhaphidophoridae) population dynamics in a mammal-free reserve
- Performed intensive night surveys and learned to identify and record over 30 different native and invasive plant species over the course of a week
- Collaborated with a team of four other students to perform statistical analyses of data and shared results via an oral presentation to community members, stakeholders, and other scientists

Summer Undergraduate Research Fellowship — University of New Hampshire, NH

June - August 2022

Undergraduate Research Fellow

- Conceptualized and wrote a research proposal regarding use of Unmanned Aerial Systems to detect and monitor Beech Bark Disease
- Performed field surveys, assessed and analyzed damage from Beech Bark Disease using ArcGIS Collector and high-precision GPS tools
- Used processing tools including ArcGIS Pro, AgiSoft MetaShape, and eCognition to develop and analyze multispectral images and orthophotos

Basic and Applied Spatial Analysis Lab — University of New Hampshire, NH

October 2021 - May 2023

Undergraduate Research Assistant

- Performed literature review research regarding the use of remote sensing and photogrammetry in grass and shrubland communities as part of a paper published in peer-review academic journal *Geographies*
- Assisted in UAS (Unmanned Aerial System) flights using both fixed-wing and quadcopter drones
- Aided fieldwork for additional projects by calculating distances and tree heights with rangefinders as well as identifying native tree species

Connors Writing Center — University of New Hampshire, NH

September 2021 - May 2023

Undergraduate Writing Assistant

- Held up to 8 one-on-one writing conferences per week with undergraduate students on projects from any curriculum
- Assisted students with structure, flow, comprehension, and grammar
- Participated in "Science Writing" committee; held discussions on how to better assist students in STEM at the Writing Center while educating non-STEM writing assistants on the conventions of scientific writing
- Participated in "Outreach" committee; visited classrooms across different curriculums to teach conventions of the peer-review process and promote the Writing Center

Student Conservation Association/U.S. Forest Service — Huron-Manistee National Forests, MI

June - August 2021

Botany and Non-Native Invasive Species Intern

- Identified and removed over 15 invasive species throughout Forest Service properties

- Assisted in mapping invasive plant communities using ArcGIS Pro
- Performed surveys of the federally threatened Pitcher's thistle and Karner Blue Butterfly

Santa Monica Mountains Fund — Santa Monica Mountains National Recreation Area, CA

February - June 2021

Restoration Program Technician

- Assisted in a variety of tasks related to the restoration program, all done virtually
- Oversaw social media work including short videos as well as weekly posts
- Participated in data input in Excel and Google Earth Pro
- Assisted in conceptualizing, planning, promoting, and hosting a virtual 500+ people conference associated with milkweed and monarch butterfly conservation

U.S. Department of the Interior — National Park Service Regional Office, Atlanta GA

September 2020 - June 2021

Virtual Student Federal Service (VSFS) eIntern

- Researched the best management practices of invasive species in national parks across the southeast, including 7 national parks and 2 U.S. territories
- Collaborated with wildlife professionals to create wildlife management protocols
- Created and researched plans to manage and count feral cat populations in San Juan, Puerto Rico through the use of camera traps

National Park Service — Santa Monica Mountains National Recreation Area, CA

November 2020 - January 2021

Social Media Intern

- Produced, filmed, scripted, and edited short promotional and educational videos for the public
- Took photographs and videos as well as wrote captions of restoration and nursery work to be used for social media posts
- Participated in restoration and native plant nursery work as well as oak tree surveys

National Park Service — Santa Monica Mountains National Recreation Area, CA

May - August 2020

Native Plant Restoration Intern

- Located, identified, and collected seed from local native plants to use for restoration and preservation efforts
- Gathered data on invasive plant species using satellite maps, GPS, and ArcGIS Field Maps
- Collected data and scouted for milkweed populations as well as utilized GPS as part of a milkweed monitoring project for monarch butterfly conservation
- Identified areas with healthy oak tree saplings by surveying areas along trails and marking points using GPS
- Assisted in general nursery tasks such as transplanting, sowing seed, and watering